Reconsidering the Lines:

Famous Reversals in Fiscal and Monetary Doctrine

Thomas J. Sargent

March 2026

Abstract

A sequel to Sargent [2011] on the trade-off between price-level stability and Ramsey-optimal tax smoothing, this essay illustrates the practical delicacy of the institutional choice between fiscal dominance and monetary dominance by cataloguing celebrated reversals. Milton Friedman crossed the divide between his 1948 proposal for coordinating monetary and fiscal policy and his 1960 program for a fixed money-growth rule. Henry Clay and James Madison each opposed the First Bank of the United States on constitutional and republican grounds, then became architects of the Second Bank once the War of 1812 had exposed the fiscal cost of having no national monetary institution. Most dramatically, Salmon P. Chase spent a career as a hard-money Jacksonian Democrat, engineered the Legal Tender Acts as Secretary of the Treasury in the Civil War, and then, as Chief Justice, wrote the opinion in *Hepburn v. Griswold* (1870) declaring his own Acts unconstitutional. The essay then turns from individual reversals to an institutional one at the congressional level: the half-century redrawing of the line between *money* and *credit*, from the bond-backed currency of the National Banking Acts that Chase championed to the real bills provisions of the Federal Reserve Act of 1913—a reversal that resolved one set of problems while embedding a different instability into the new system. These reversals are not signs of weakness or inconsistency; they are evidence of how contextually sensitive, and how consequential, the institutional lines really are.

Contents

1 Introduction **3**

2 The Framework Recalled **4**

3 Milton Friedman: From Fiscal Coordination to Monetary Rules **5**

4 Henry Clay: From Republican Critic to Champion of the American System **7**

5 James Madison: Constitutional Principle and the Doctrine of Liquidation **9**

6 Salmon P. Chase: From Hard Money to Legal Tender to Constitutional Revision **11**

6.1 A Witness of Singular Authority 11
6.2 The Hard-Money Democrat 12
6.3 The Legal Tender Acts and the 5-20 Bonds 13
6.4 *Hepburn v. Griswold* (1870): Chase's First Reversal 14
6.5 *Knox v. Lee* (1871): The Court-Packing Reversal 15
6.6 *Juilliard v. Greenman* (1884): Legal Tender as Permanent Sovereignty 16
6.7 Constitutional Tragedy and Soft Constraints 18

7 The Money–Credit Line: From Bond-Backing to Real Bills, 1863–1913 **19**

7.1 Bond-Backing and the Line at Government Debt 19
7.2 The Inelasticity Debate as Four Decades of Disagreement about the Line 20
7.3 The Real Bills Resolution and Its Costs 21
7.4 The Theoretical Afterlife and the Sargent–Wallace Reconsideration . 22

8 Common Themes **23**

9 Conclusion **24**

1 Introduction

Where to draw the institutional boundary between monetary and fiscal authority is one of the deepest questions in macroeconomic design. Sargent [2011] argued that this question poses a genuine dilemma: arrangements that support price-level stability tend to foreclose the Ramsey-optimal tax smoothing that Robert E. Lucas and Stokey [1983] showed requires using money creation as a flexible fiscal instrument, while arrangements that enable efficient public finance tend to inject fiscal pressures into monetary policy and thereby risk inflation. There is no dominant option. The choice depends on the fiscal environment, the quality of monetary institutions, the credibility of commitments, and the weight placed on different social objectives.

If the choice were straightforward, one would expect thoughtful and principled people to make it once, correctly, and stick to it. The historical record suggests otherwise. Some of the most able statesmen and economists in American history changed their minds, sometimes more than once, and sometimes in ways that placed them in direct tension with their earlier selves. This essay examines four such individual reversals—Friedman, Clay, Madison, Chase—and then turns to a fifth reversal that unfolded at the institutional level over half a century: Congress's stepwise redrawing of the line between money and credit, from the bond-backed National Bank Notes of the 1860s to the real-bills-backed Federal Reserve notes of 1913.

Milton Friedman, who serves as the primary example in Sargent [2011], moved from a 1948 proposal that coordinated money creation tightly with fiscal deficits—an arrangement that prioritizes efficient stabilization but makes monetary policy subordinate to budget outcomes—to the 1960 program for a fixed k-percent money-growth rule that insulates money entirely from the budget, forfeiting tax-smoothing efficiency in the name of price-level stability. Intellectual development accounts for part of Friedman's shift; but the shift also reflects a judgment that the institutional capacity for responsible fiscal coordination had been permanently compromised by Keynesian intellectual fashions and political incentives.

Henry Clay and James Madison present a complementary pattern at an earlier moment in American history. Both opposed the First Bank of the United States—Clay as a young Jeffersonian congressman on republican and states'-rights grounds, Madison as the leading constitutional theorist of the founding generation. The financial

catastrophe of the War of 1812, fought without a national bank capable of mobilizing government credit, chastened them both. Clay would make a national bank the centerpiece of his"American System"; Madison would sign the bill creating the Second Bank in 1816 having argued with new sophistication that long legislative practice had itself resolved the constitutional question.

The most dramatic reversal belongs to Salmon P. Chase. A hard-money Jacksonian Democrat whose early career was defined by opposition to paper currency and to the mixing of public and private credit, Chase became Abraham Lincoln's Secretary of the Treasury in 1861 and proceeded to design and defend the Legal Tender Acts of 1862 and 1863, which made inconvertible paper notes ($450 million in "greenbacks") legal tender for all debts. Appointed Chief Justice in 1864, Chase spent the rest of his career trying to undo his own handiwork. In *Hepburn v. Griswold* (1870) he wrote the majority opinion holding the Legal Tender Acts unconstitutional as applied to debts contracted before their passage. Grant's addition of two new justices reversed that decision in *Knox v. Lee* (1871). Chase dissented.

The argument is not that these figures were confused or hypocritical. Each reversal was accompanied by sustained reasoning and, usually, by an explicit account of why changed circumstances or changed understanding warranted a different institutional line. What the reversals collectively demonstrate is that the stability-versus-efficiency trade-off identified in Sargent [2011] is not merely a theoretical curiosity; it maps onto live political and constitutional choices that intelligent people answer differently in different contexts, and sometimes answer differently over the course of a single lifetime.

2 The Framework Recalled

Sargent [2011] builds on the Sargent and Wallace [1981] "unpleasant monetarist arithmetic" framework. Let the government's flow budget constraint be

$$b_t + \tau_t + \mu_t m_{t-1} = g_t + (1 + r_{t-1})\, b_{t-1}, \tag{1}$$

where b_t is real debt, τ_t real tax revenues, g_t real spending, r_{t-1} the real interest rate on debt, and $\mu_t = (M_t/M_{t-1} - 1)$ the money-growth rate applied to a real money base

m_{t-1}. Seigniorage $\mu_t m_{t-1}$ enters as a source of revenue that competes with ordinary taxation.

Two polar institutional arrangements are possible:

Monetary dominance. The monetary authority commits credibly to a path for $\{M_t\}$ (or equivalently the price level), and the fiscal authority must choose $\{\tau_t,\ b_t,\ g_t\}$ to satisfy (1) without altering that path. Fiscal policy is the "residual claimant"; seigniorage is fixed exogenously by the monetary rule. Price-level stability is guaranteed by the commitment, but tax-smoothing efficiency is constrained: the fiscal authority cannot use seigniorage as a flexible instrument to smooth distortionary taxes across states and dates.

Fiscal dominance. The fiscal authority sets $\{\tau_t,\ g_t,\ b_t\}$ autonomously and the monetary authority passively accommodates whatever seigniorage is required to clear (1). Ramsey-optimal tax smoothing is now achievable: if shocks to g_t are sufficiently persistent, Robert E. Lucas and Stokey [1983] show that optimality calls for using money creation alongside other taxes. But fiscal dominance makes the price level a fiscal residual; any deterioration in fiscal discipline is monetized and price-level instability follows.

The core result of Sargent [2011] is that neither arrangement Pareto dominates the other. Monetary dominance trades efficiency for stability; fiscal dominance trades stability for efficiency. The social ranking depends on parameterization and context. This is the sense in which the "line" is not obvious: reasonable people, reasoning carefully, can end up on different sides depending on their assessment of the relevant fiscal and monetary environment. The historical reversals documented below are vivid illustrations.

3 Milton Friedman: From Fiscal Coordination to Monetary Rules

Friedman's 1948 paper in the *American Economic Review*, "A Monetary and Fiscal Framework for Economic Stability" [Friedman, 1948], proposed an institutional architecture in which monetary policy was explicitly subordinated to fiscal outcomes. The proposal rested on three pillars: (1) a requirement that commercial banks hold 100

percent reserves against deposits, so that money creation was entirely a government function; (2) automatic fiscal stabilizers (progressive taxation and transfer payments) that would cause the budget to move into surplus during booms and deficit during recessions without discretionary action; and (3) a rule specifying that any deficit be financed exclusively by issuing money, while any surplus be used to retire money. Under this system, the supply of money was mechanically linked to the fiscal balance: the government literally printed money to finance shortfalls and destroyed money when revenues exceeded expenditures.

This is a case of programmatic fiscal dominance. The monetary instrument—the money supply—is not controlled by an independent central bank pursuing a price-level target; it is set residually to satisfy a fiscal accounting identity. In the terminology of Sargent [2011], the monetary authority has been dissolved into the fiscal authority. Friedman embraced this design in 1948 because he believed that the economic cycle was predominantly driven by changes in aggregate demand, that discretionary monetary and fiscal management were dangerously unreliable, and that the automatic, formula-driven coupling of money creation to fiscal outcomes would deliver macroeconomic stabilization without giving policymakers enough discretion to do harm. His touchstone at this stage was the *coordination* of the two policy arms, not their separation.

By the late 1950s Friedman had moved decisively. His Fordham University lectures, published as *A Program for Monetary Stability* [Friedman, 1960], advocated for a fixed k-percent annual growth rate in a well-defined monetary aggregate, entirely independent of fiscal conditions. The Federal Reserve should expand the money supply at a constant rate—Friedman suggested roughly 3 to 5 percent per year—regardless of whether the budget was in surplus or deficit. This is a textbook case of monetary dominance: the monetary rule is the binding commitment, and the fiscal authority must accommodate it.

The intellectual distance between the two positions is large. Under the 1948 framework, a persistent fiscal deficit would automatically generate money creation and hence inflation; the 1960 framework explicitly blocks this channel. What changed?

Friedman identified several reasons [Friedman, 1960], Chapter 4. First, he had become convinced by the historical evidence, which he and Friedman and Schwartz [1963] were

then assembling, that autonomous monetary shocks—changes in the money stock not driven by fiscal conditions—were the dominant source of macroeconomic fluctuations. The Great Depression was not a collapse of aggregate demand correctable by fiscal-monetary coordination; it was a catastrophic monetary contraction engineered by Federal Reserve mistakes. If monetary shocks were the main danger, the right institutional response was to bind the monetary authority, not to couple it to fiscal outcomes.

Second, Friedman had concluded that fiscal discipline in a democratic polity was essentially unenforceable. Governments that could finance deficits by money creation would reliably do so. The 1948 design assumed that the fiscal stabilizers would be genuinely automatic and symmetric: surpluses in good times would destroy money just as deficits in bad times created it. But the historical record suggested that democracies enlarged their fiscal commitments during downturns and failed to retire them during expansions. A design that coupled monetary policy to fiscal outcomes would therefore degenerate into a one-way ratchet toward inflation.

Third, and most analytically interesting from the perspective of Sargent [2011], Friedman had implicitly revised his assessment of the stability-efficiency trade-off. The 1948 framework prized efficiency (smooth automatic stabilization) and accepted the price-level risk entailed by fiscal coupling. The 1960 program prized stability (a committed monetary rule) and accepted the efficiency loss from foreclosing flexible seigniorage. The change was not a logical correction of an earlier error; it was a shift in the relative weight placed on the two objectives, driven by an updated reading of what the American political economy could actually sustain.

Friedman's reversal is a clean illustration of the paper's main thesis: the optimal institutional line between fiscal and monetary authority depends on empirical and institutional features of the environment that are themselves changing and contestable.

4 Henry Clay: From Republican Critic to Champion of the American System

Henry Clay entered Congress as a Kentucky senator in 1806, steeped in the Jeffersonian Republican tradition that regarded the First Bank of the United States as an

unconstitutional concentration of financial power in private hands. When the First Bank's twenty-year charter expired in 1811 and recharter was debated, Clay—now a newly elected member of the House—was among those who voted against it. The recharter bill died narrowly: the Senate divided 17–17, and Vice President George Clinton cast the deciding vote against; the House defeated it 65 to 64. Clay's opposition was rooted in the Jeffersonian political economy that Andrew Jackson would later systematize: the Bank was a creature of Hamilton's Federalist designs, an engine of privilege and corruption, constitutionally suspect, and dangerous to the democratic republic.

Five years later, Clay was one of the most forceful advocates in Congress for creating the Second Bank of the United States. The Second Bank was chartered in April 1816 with a twenty-year term, capitalized at $35 million (more than three times the First Bank's capitalization), and endowed with explicit authority to issue notes that could serve as currency nationwide. Clay's support was enthusiastic and, more importantly, principled: he incorporated the national bank into his "American System," the integrated program of protective tariffs, federally financed internal improvements, and a national monetary institution that he argued was necessary for American industrial development, national security, and economic integration.

What happened between 1811 and 1816 was the War of 1812. In the absence of a national bank, the Madison administration found it nearly impossible to finance the war. State banks proliferated and issued paper with no common standard; the government was reduced to borrowing at punishing terms from private financiers; military operations were hampered by the Treasury's inability to transfer funds efficiently across the continent. In 1814, state banks outside New England suspended specie payments—an event that dramatized the instability of a purely decentralized monetary system. The government's own fiscal operations were threatened: Treasury notes circulated at steep discounts, and the Secretary of the Treasury reported that he could not reliably meet the government's payroll.

Clay's reversal is intelligible as a response to this evidence. Before the War of 1812, the costs of a national monetary institution—the constitutional objections, the concentration of financial privilege, the threat to state banking interests—loomed large relative to its benefits, which seemed speculative and ideological. After the war, the corresponding cost-benefit calculation ran the other way: the demonstrated fiscal cost

of the absence of a national institution was concrete and severe, while the constitutional objections had been blunted by the practical success of the First Bank during its twenty years of operation. The line between efficient national finance and the dangers of centralized monetary power had moved, not because the underlying principle had changed, but because the evidence on the costs of each institutional arrangement had accumulated.

In the framework of Sargent [2011], the First Bank had been an instrument of fiscal-monetary coordination: it held government deposits, facilitated tax collection, managed government debt, and provided a measure of monetary discipline through its willingness to redeem its notes. The two decades after its expiration amounted to an unplanned experiment in the absence of such coordination. Clay read the experimental results and updated accordingly. That is not inconsistency; that is rationality.

The sequel to Clay's conversion is worth noting. When Andrew Jackson vetoed the recharter of the Second Bank in 1832 and destroyed it by withdrawing government deposits in 1833–34, Clay emerged as the leading opponent of Jackson's action. The Bank War became the defining issue of Clay's Whig Party. The man who had voted against the First Bank in 1811 became, by the 1830s, the foremost defender of the principle of a national monetary institution against Jacksonian hard-money democracy. The reversal had become the fixed point from which he did not thereafter move.

5 James Madison: Constitutional Principle and the Doctrine of Liquidation

James Madison's trajectory runs parallel to Clay's and, in some respects, deeper, because Madison's original opposition to the First Bank was not merely political but philosophical—grounded in the most careful constitutional theory of his generation.

When Alexander Hamilton proposed the First Bank in January 1791, Madison led the House opposition. His speech of February 2, 1791 is one of the classic statements of strict constructionism. The Constitution granted Congress enumerated powers. The power to charter a corporation was not among them. The "necessary and proper" clause could not be interpreted to extend congressional power to any means that might

be deemed convenient or useful; "necessary" had to mean genuinely indispensable to the execution of an enumerated power, and chartering a bank was not indispensable to anything the Constitution explicitly authorized. Madison also invoked what would later be called the principle of original understanding: the framers' intentions, as manifested in the ratification debates, gave no support for so broad an implied power.

Hamilton's counter-argument won: Washington accepted Hamilton's opinion and signed the bill. But Madison's position was not merely a tactical loss; it established the intellectual framework that Jeffersonian Republicans would invoke for the next two decades, and that Jackson's Democrats would recycle in the Bank War of the 1830s.

In 1815, near the end of the war and his presidency, Madison vetoed a bank bill—not on constitutional grounds, but because the proposed bank lacked features that would make it an effective fiscal instrument for the government. The veto message explicitly declined to reopen the constitutional question, which Madison said had been settled by congressional and executive practice over twenty-four years. The following year, in 1816, Madison signed the bill creating the Second Bank of the United States.

How did the author of the 1791 speech against bank constitutionality come to sign the bill creating a larger successor? Madison's answer was sophisticated and, to his critics, infuriatingly convenient. He had articulated it already in the 1815 veto message: while his own constitutional views had not changed, the Constitution's meaning was not fixed solely by the original understanding of individual framers. It was also determined by the accumulated legislative, executive, and judicial practice of the republic. A course of action repeatedly approved by Congress, acquiesced in by successive presidents, and never invalidated by the courts thereby acquired constitutional legitimacy through what Madison called the process of "liquidation"—the settlement of constitutional meaning by institutional practice over time. By 1816, twenty-four years of the First Bank's operation had, in Madison's view, liquidated the constitutional question in favor of congressional authority.

The liquidation doctrine is philosophically consequential. It implies that the institutional line between permissible and impermissible concentration of fiscal-monetary power is not fixed by original text alone; it evolves through a common-law-like process of precedent and practice. Institutions that were once constitutionally contested can

become constitutionally settled, and the initial objections correspondingly lose their force. In the terms of Sargent [2011], the boundary between stability-promoting and efficiency-promoting arrangements is not only analytically ambiguous; it is constitutionally labile, subject to reinterpretation as practice accumulates.

Madison's reversal also reflects the same lesson Clay drew from 1812: the War had revealed the fiscal consequences of the absence of a national monetary institution. A republic that could not finance its wars without submitting to ruinous private financiers was not well served by constitutional principles, however impeccably derived. The stability of the polity—its fiscal survival—was itself at stake; abstract monetary-constitutional purity had proved a luxury that the republic could not afford.

The most important contrast to Madison is his friend and rival Thomas Jefferson, who never accepted the liquidation doctrine and to the end of his life insisted that the Bank was unconstitutional. Jefferson's consistency and Madison's flexibility represent the two sides of what is today still a live debate: whether institutional practice can and should revise constitutional meaning, or whether original principle must constrain practice however inconvenient the results. That debate is, in the end, about who gets to move the lines—and whether the lines can legitimately be moved at all.

6 Salmon P. Chase: From Hard Money to Legal Tender to Constitutional Revision

The most internally dramatic of the reversals documented here belongs to Salmon P. Chase. His career traces a complete arc: from principled hard-money doctrine, through the coercion of wartime necessity, to a judicial crusade to undo his own executive work. Few episodes in American constitutional history have been dissected more searchingly than this one. The fullest contemporary anatomy is George Bancroft's *A Plea for the Constitution of the United States Wounded in the House of Its Guardians* [Bancroft, 1886], a remarkable document that repays study here.

6.1 A Witness of Singular Authority

Bancroft (1800–1891) was uniquely positioned to indict the Legal Tender Cases. As the foremost narrative historian of the United States—his ten-volume *History of the*

United States (1834–74) was the monumental scholarly achievement of the century—he brought to monetary constitutionalism not merely the moralism of a hard-money partisan but the documentary authority of a man who had spent fifty years in the primary sources of the founding era. He had personally known James Madison and John Adams; had served as Secretary of the Navy under Polk and as Minister to Prussia and Germany under Johnson and Grant; and was 84 years old when, in the wake of *Juilliard v. Greenman* (1884), he sat down to write his *Plea*. The book is a comprehensive historical argument that the framers at Philadelphia had deliberately and by large majorities stripped from the federal government any power to make paper money legal tender for private debts—and that a sequence of Supreme Court decisions had betrayed that design. His central exhibit, running through every part of the work, is Salmon P. Chase.

Bancroft's constitutional argument rests on what happened in Philadelphia on August 16, 1787. The original draft of the Constitution had given Congress the power "to borrow money, and emit bills on the credit of the United States." Gouverneur Morris moved to strike the words "and emit bills." The motion passed 9–2. Madison's convention notes recorded that the majority "thought it advisable not to insert this power; thinking it probable that the omission would not disable the use of public notes so far as they would be safe and proper, and would only cut off the pretext for a paper currency, and particularly for making the bills a tender either for public or private debts." Article I, Section 10 then explicitly forbade the states from emitting bills of credit or making anything but gold and silver coin legal tender. For Bancroft, this two-sided design—the power denied to Congress by omission, the power denied to the states by express prohibition—was the clearest possible evidence that the framers had closed the door to paper legal tender at every level of government.

Chase's career is, in Bancroft's telling, the story of that door being pried open and the constitution bleeding as a result.

6.2 The Hard-Money Democrat

Chase began his public career in the 1830s and 1840s embedded in the Jacksonian political tradition. The Jacksonian hard-money creed, which Chase shared, held that paper money not convertible to specie was a form of fraud on holders and debtors alike; that the issuance of bank notes by private institutions constituted an illegitimate

privilege; and that sound money meant gold and silver coin, or notes immediately and unconditionally redeemable in them. Chase was a Free Soiler and then a Republican rather than a Democrat by the 1850s, but his monetary convictions retained their Jacksonian character: he distrusted paper, trusted specie, and consistently argued that mixing governmental and banking functions corrupted both.

As Governor of Ohio (1856–60) and then Senator, Chase never departed from these positions. When Lincoln appointed him Secretary of the Treasury in March 1861, Chase brought this monetary philosophy with him. His first instinct was to finance the Civil War through taxation and borrowing at market terms, without resort to inconvertible paper. He raised tariffs, promoted the new income tax, and placed bond issues with Jay Cooke and other financial houses. In his initial communications to Congress, Chase explicitly rejected proposals for a legal-tender paper currency.

6.3 The Legal Tender Acts and the 5-20 Bonds

By December 1861 the financial situation had become critical. Banks had suspended specie payments; the Treasury could not meet its obligations; military expenditures were running at rates that bond markets could not absorb at sustainable interest rates. Chase reversed course completely. He endorsed, designed, and lobbied vigorously for the Legal Tender Act of February 25, 1862, which authorized the issuance of $150 million in Treasury notes—"United States Notes," quickly styled "greenbacks"—that were legal tender for all debts, public and private, but not convertible to specie. Subsequent acts in July 1862 and March 1863 authorized additional issues, bringing the total to approximately $450 million.

Alongside the greenbacks, Chase issued the so-called 5-20 bonds to finance the war: six-percent coupon bonds redeemable after five and payable within twenty years, whose authorizing statute (the Act of February 25, 1862) specified that interest would be paid in coin but was entirely silent on principal. Chase bridged this statutory gap with reputational commitment, marketing the bonds to the public—through Jay Cooke's enormous retail campaign—with the implicit understanding that principal, too, would eventually be repaid in gold. The commitment was extra-statutory: it rested on Chase's personal credibility as a hard-money man, not on any legislative guarantee. That would matter enormously later.

The arguments Chase offered for the Acts were purely fiscal and pragmatic: the government faced a choice between legal tender paper and either national bankruptcy or negotiated independence. He invoked the precedent of the Revolutionary War and argued that the power to issue legal-tender notes was inherent in sovereignty and necessary to the government's most vital function—its survival in armed conflict. He did not pretend that this was consistent with his earlier views; he argued only that emergency had foreclosed the options he would have preferred.

Bancroft does not excuse Chase for this. The Revolutionary precedent was, for Bancroft, precisely the wrong one to invoke: continental dollars were the most instructive episode in all of American monetary history, the reductio ad absurdum of paper money, and the primary inspiration for the Philadelphia framers' decision to close the door. To cite the Revolutionary experience as precedent was to cite the disease as a justification for a recurrence.

In the taxonomy of Sargent [2011], the Legal Tender Acts were a decisive move toward fiscal dominance. The government issued an inconvertible currency and declared it legal tender; the "monetary policy" embedded in this decision—the price level would be whatever was required to make $450 million in greenbacks cover real government expenditures—was entirely subordinate to fiscal necessity. Chase had, under pressure of circumstances, crossed from the hard-money extreme of monetary dominance (specie standard, no fiat paper) to the fiscal-dominance extreme (inconvertible legal-tender paper issued at the government's discretion).

6.4 *Hepburn v. Griswold* (1870): Chase's First Reversal

Lincoln appointed Chase Chief Justice in December 1864. For the remainder of his life, Chase pursued a constitutional agenda centered on reversing the monetary consequences of his own Treasury policy.

The occasion came in *Hepburn v. Griswold* (75 U.S. 603), decided February 7, 1870. The case involved a debt of $11,250 contracted in 1860, before the Legal Tender Act, and tendered for payment in 1864 with greenbacks. The creditor, Mrs. Hepburn, argued that she was entitled to gold coin equivalent, since the contract antedated the Act that made paper legal tender. By a 4–3 vote (two seats were vacant), the Court held for the creditor. Chase wrote the majority opinion.

The *Hepburn* opinion rested on two grounds. First, Chase argued that compelling creditors to accept depreciated paper for pre-existing gold debts deprived them of property without due process in violation of the Fifth Amendment. Second, he argued that making greenbacks legal tender for such debts was not within the Necessary and Proper Clause: Congress could permissibly issue paper currency to finance the war, but the further step of compelling private creditors to accept it for pre-existing obligations was not "necessary" to any express power in the required sense. Crucially, Chase appealed not to express textual prohibition but to the "spirit" of the Constitution—a methodological choice that would shortly be turned against him.

The opinion conspicuously did not address whether Chase had been correct in 1862 to recommend the Acts. But the logical entailment was inescapable: if making greenbacks legal tender for pre-existing debts was unconstitutional, then Chase had committed a constitutional error as Secretary. Justice Samuel Miller's dissent noted quietly, and devastatingly, that the majority had struck down legislation that the Chief Justice himself had recommended and defended as a member of the executive branch. Miller also criticized Chase's appeal to the "spirit" of the Constitution as "abstract and intangible" reasoning that invited an unlimited judicial veto over congressional action.

6.5 *Knox v. Lee* (1871): The Court-Packing Reversal

The reversal of *Hepburn* came with stunning speed and in circumstances whose constitutional impropriety has been debated ever since. On the very day *Hepburn* was announced—February 7, 1870—President Grant nominated two new justices to fill existing vacancies: William Strong of Pennsylvania and Joseph Bradley of New Jersey. Chase believed, and said, that Grant had deliberately timed the nominations to reconstitute the Court and obtain a reversal. When the Court reached its new complement of nine, the Attorney General moved for reargument of the legal tender question. *Knox v. Lee* and *Parker v. Davis* (79 U.S. 457) were decided together on May 1, 1871, by a vote of 5–4, overruling *Hepburn*.

Justice Strong's majority opinion relied on the implied-powers doctrine of *McCulloch v. Maryland*: if Congress has the war power and the power to borrow money, and if a legal-tender paper currency is an appropriate means of executing those powers, the Acts are constitutional under the Necessary and Proper Clause. Strong's opinion

included a remarkable passage—remarkable even by the standards of intra-tribunal comity—chiding Chase for having "forced" an earlier decision when the Court was divided and on the verge of receiving new appointees. As Bancroft observed, this was the majority lecturing the former majority about the indecorum of the very process by which the current majority had come to exist.

Chase dissented, joined by Justices Clifford and Field. His dissent insisted that the majority had abandoned the constitutional limitations on congressional power in favor of an open-ended theory of implied powers that left no meaningful constraint on federal action. Field's dissent was the most constitutionally principled of the three: the doctrine that whatever is not expressly forbidden may be exercised would "change the whole character of our government" from one of limited, enumerated powers to one of unlimited powers in its relations to the people.

Bancroft regarded the Court's composition question as itself corroborating his thesis. On a constitutional question of the first magnitude—whether the federal government may compel private citizens to accept depreciated paper in discharge of gold obligations—the outcome turned not on any change in the law or the facts but on the personnel of the Court. This was, in constitutional terms, not a "hard constraint" (in the sense of a textual prohibition too clear to circumvent) but a "soft constraint" that yielded to political pressure once the political composition of the appointing authority changed. The same logic that had made Chase's reputational commitment on the 5-20 bond principal extra-statutory and therefore fragile had made the constitutional prohibition on legal tender extra-textual and therefore fragile.

6.6 *Juilliard v. Greenman* (1884): Legal Tender as Permanent Sovereignty

The third and final case swept away whatever remained of the *Hepburn* principle. In *Juilliard v. Greenman* (110 U.S. 421), decided March 3, 1884, Justice Gray's majority opinion—joined by eight of nine justices—held that the power to declare paper money legal tender was not limited to wartime emergencies but was a permanent attribute of national sovereignty. The statutory notes at issue had originally been issued during the war, then redeemed and reissued in peacetime under the Resumption and Refunding Act of 1878; no emergency justified them. That did not matter. Gray

wrote:

> The power to make the notes of the government a legal tender in payment of private debts being one of the powers belonging to sovereignty in other civilized nations, and not expressly withheld from congress by the constitution; we are irresistibly impelled to the conclusion that the impressing upon the treasury notes of the United States the quality of being a legal tender in payment of private debts is an appropriate means, conducive and plainly adapted to the execution of the undoubted powers of congress.

For Bancroft, this language was "the language of revolution." The argument from what "other civilized nations" had done had no place in interpreting a constitution deliberately designed to withhold powers that every European monarchy exercised. The argument from "not expressly withheld" converted a government of enumerated powers into one of unlimited powers: if Congress may do anything not expressly forbidden, then the framers' painstaking enumeration of powers was a dead letter. Moreover, the same nine justices who joined the *Juilliard* majority had unanimously declared, just two years earlier, that "the government of the United States is one of delegated, limited, and enumerated powers," and that "every valid act of congress must find in the constitution some warrant for its passage." Bancroft noted acidly that the *Juilliard* opinion was therefore "in flagrant antagonism" to the Court's own recent, unanimous, and repeatedly given interpretations of the constitutional structure.

The sole dissenter was Justice Field—who had been in Chase's *Hepburn* majority and in the *Knox* dissent, and who now stood entirely alone. Bancroft's rhetorical move was to insist that Field was not a minority of one but the transmitted judgment of the founding generation: standing beside him were "invincible vouchers for the rightness of his dissent"—James Wilson, Oliver Ellsworth, and William Paterson, all three appointed to the Court by Washington from among the Convention's own framers. The institutional line that the framers had drawn at Philadelphia, that Chase had tried to restore from the bench, and that Field had defended alone, had been entirely erased.

6.7 Constitutional Tragedy and Soft Constraints

Bancroft did not treat Chase's trajectory with personal relish. He read it as symptomatic of a constitutional culture that had lost its moorings rather than as individual hypocrisy. But his framing makes the theoretical point with unusual clarity. Chase's reputational commitment to gold repayment of the 5-20 bond principal and his constitutional commitment to the invalidity of legal-tender paper were both *extra-statutory*—both existed as assurances given without explicit legislative or textual authority. Both were undone by the same political logic that had originally forced Chase to accept the Legal Tender Acts as a war measure. When the political environment changed enough to matter, neither commitment was self-enforcing.

For the purposes of Sargent [2011], the arc of Chase's career encapsulates the full range of the stability–efficiency trade-off. His early Jacksonian position was an extreme form of monetary dominance: specie only, no government paper, the fiscal authority must adjust entirely to monetary reality. The Legal Tender Acts represented a wartime capitulation to fiscal dominance: the monetary instrument becomes whatever fiscal survival requires. His judicial career was an attempt to restore monetary dominance by constitutional command—to fix in law the institutional line against which executive necessity, however acute, could not encroach.

The *Hepburn–Knox–Juilliard* trilogy demonstrates that the attempt failed, and *why* it failed. The constitutional constraint Chase sought to enforce was a soft constraint: it rested not on language expressly prohibiting the legal tender power but on the "spirit" of the Constitution as Chase interpreted it, and on the historical evidence that the Philadelphia Convention had deliberately struck out the "emit bills" clause. Bancroft thought this soft constraint should have been treated as a hard one; the Court disagreed, and the political environment— unwilling to foreclose a fiscal instrument it had used to win a civil war—did not sustain the harder reading. Institutional lines that are not politically sustainable cannot be maintained by constitutional command alone, and Chase's career is the clearest proof.

7 The Money–Credit Line: From Bond-Backing to Real Bills, 1863–1913

The reversals catalogued above all concern the boundary between the monetary and fiscal *authorities*: who controls the supply of money, and whether the fiscal authority may commandeer the monetary instrument. But there is a related and equally contested institutional line—the line between *money* and *credit*—and the half- century following Chase's National Bank Acts is a case study in Congress repeatedly redrawing it.

7.1 Bond-Backing and the Line at Government Debt

The National Bank Act of 1864 drew the money–credit line at an unusual location: United States government bonds. National Bank Notes—the dominant hand-to-hand currency of the post-Civil War era—could be issued only against specified federal bonds deposited with the Comptroller of the Currency. That institutional officer, whose primary function was verifying the bond portfolios, could certify the soundness of every dollar in circulation. The line between what counted as "money-eligible" and what counted as mere "credit" was the difference between government bonds on one side and every other asset—commercial loans, railroad securities, warehouse receipts, farm mortgages—on the other.

In the framework of Sargent [2011] this arrangement is a form of *monetary dominance*, but with a peculiar twist: the binding constraint on the money supply was not a target set by an independent central bank but the stock of outstanding federal debt. When the government ran the large primary surpluses of the 1870s and 1880s and retired war bonds, the supply of eligible collateral contracted, and the currency supply contracted with it regardless of any commercial demand for more money. Fiscal virtue, no less than fiscal profligacy, could thus distort the monetary instrument; the two distorted it in opposite directions, but neither direction was chosen for monetary reasons. The bond-backing requirement was a hard constraint against inflationary over-issue, but it purchased that stability by tying the money supply to the government's debt-management calendar rather than to the needs of trade.

7.2 The Inelasticity Debate as Four Decades of Disagreement about the Line

Critics from the 1870s onward called the resulting currency "inelastic." The agrarian South and West needed sharply higher volumes of currency during planting and harvest seasons; the bond-backed system offered no mechanism for such expansion. Financial panics punctuated the era with depressing regularity. The political response produced two broad coalitions, each proposing to move the money–credit line in a different direction.

Greenbackers and silver inflationists wanted to dissolve the line almost entirely. If the government simply issued more fiat currency—greenbacks to be increased rather than retired, or silver to be coined at a generous ratio—there was no need for any specific asset to "back" the money at all. This was, at the limit, the same fiscal dominance that Chase had temporarily embraced during the war: the money supply would be whatever fiscal or political convenience required. William Jennings Bryan's "Cross of Gold" speech of 1896 crystallized this position in its most politically potent form.

Asset-currency reformers drew the line elsewhere. Their complaint was not too little money but the wrong kind of backing. They argued that national banks should be permitted to issue notes against a wider range of productive assets—railroad and corporate bonds, agricultural warehouse receipts, or any "good assets" on a bank's balance sheet—rather than against government bonds alone. This was a proposal to relocate the money–credit boundary from "government debt only" to "any sufficiently sound private claim." By broadening the eligibility set, they hoped, the currency supply would be free to expand with productive activity rather than with the Treasury's debt schedule.

The Aldrich–Vreeland Act of 1908, passed in the wake of the Panic of 1907, was a temporary halfway house. It allowed groups of national banks to issue emergency currency backed by "almost any securities the banks were holding," but imposed a punitive tax schedule to ensure rapid retirement once conditions normalized. More consequentially, it created the National Monetary Commission, chaired by Senator Aldrich, whose multi-volume studies of European central banking provided the intellectual scaffolding for the Federal Reserve Act of 1913.

7.3 The Real Bills Resolution and Its Costs

The Federal Reserve Act resolved the inelasticity debate by drawing the money–credit line at a new location: short-term, self-liquidating commercial paper. Section 13 of the Act confined Federal Reserve discount operations to notes and bills "arising out of actual commercial transactions" maturing within ninety days, and explicitly excluded paper drawn to finance positions in "stocks, bonds, or other investment securities." The so-called *real bills doctrine*— also called the commercial loan theory of banking, with roots in Adam Smith's *Wealth of Nations* and a long subsequent history—became the operating principle of the new institution.

The new line distinguished money from credit not by asset class but by the *purpose and duration* of the underlying transaction. A sixty-day bill drawn against a shipment of grain was eligible because the transaction was real, productive, and self-liquidating: when the grain was sold, the bill would be repaid and the money supply would contract automatically. A call loan financing stock market speculation was ineligible. In the political rhetoric of the era the line ran between Main Street and Wall Street; in the theoretical rhetoric it ran between "money backed by real production" and "money backed by finance."

The real bills criterion solved the inelasticity problem: the money supply could now expand with productive activity rather than with the stock of federal debt. But it imported a different instability, one that Henry Thornton had identified as early as 1802 in his *Paper Credit of Great Britain* [Thornton, 1802]. By accommodating every credit demand arising from commercial transactions valued at *current prices*, the criterion set up a positive feedback from prices to money creation to prices. In a boom, prices rise; the nominal value of commercial transactions rises; the demand for discount rises; and the Fed, faithfully applying the real bills criterion, expands the money supply—adding fuel to the expansion rather than restraining it. The criterion is inherently procyclical.

The most consequential application came in 1929–1933. The Federal Reserve Board member Adolph Miller launched a "Direct Pressure" policy: any bank seeking discount window relief had to certify that it had never made speculative loans. Banks, unwilling to submit to the implied interrogation, refrained from borrowing reserves even in the face of withering deposit outflows. Instead of discounting at the window,

banks failed—nearly 9,000 of them. Friedman and Schwartz [1963] held the resulting one-third contraction of the money stock largely responsible for the Great Depression. Where the bond-backed currency of the national banking era had been passively contractionary in the face of fiscal surpluses, the real bills currency proved passively contractionary in the face of banking panics. Each line carried its own characteristic failure mode.

7.4 The Theoretical Afterlife and the Sargent–Wallace Reconsideration

The real bills debate connects directly to the framework of Sargent [2011]. In a 1982 article, Sargent and Wallace [1982] showed that within overlapping-generations models the real bills prescription—unfettered private intermediation against productive assets—can be Pareto superior to the quantity-theory alternative, which imposes restrictions on intermediation in order to control the price level. This is a version of the stability–efficiency trade-off that Sargent [2011] identifies in the fiscal–monetary domain: the quantity-theory regime purchases price-level stability at the cost of constrained intermediation; the real bills regime allows efficient intermediation at the cost of weaker price-level control. Laidler [1984] objected that the Sargent–Wallace version had abstracted away the key feature of the historical doctrine—the restriction to specific collateral types—but the theorem demonstrated that the money–credit distinction is not merely an institutional accident or a historical curiosity. It reflects a genuine analytical trade-off whose terms depend on the structure of the economy and the quality of the regulatory environment.

The congressional progression from bond-backed currency to real bills—from the National Bank Acts of 1863–64 to the Federal Reserve Act of 1913, through four decades of inelasticity complaint and asset-currency agitation—thus illustrates in miniature the same dynamic as the individual reversals catalogued in this paper. The actors who drafted the Federal Reserve Act were consciously reversing the institutional design that Chase had supervised as Secretary of the Treasury. Their reversal solved one class of problems while creating another; and the real bills doctrine embedded in the new Act would itself require substantial revision in the Banking Acts of 1932–35, a revision precipitated by the very failure mode that Thornton had foreseen in 1802. The half-century arc from National Bank Notes to Federal Reserve notes is another

turn of the same wheel: each new institutional line is drawn in response to the failures of the previous one, and each generates its own successor crisis.

8 Common Themes

Several themes recur across these reversals.

War as a forcing function. Three of the four reversals were precipitated by war or its aftermath: the War of 1812 converted Clay and Madison; the Civil War broke Chase. War generates fiscal demands that peaceful institutions are designed to defer or distribute across time; when those demands are concentrated and immediate, they overwhelm standing institutional arrangements. The fiscal-dominance arrangement—using monetary instruments to finance emergency expenditures—becomes irresistible precisely because the alternative is defeat. This is not an argument against institutional constraints on fiscal-monetary coordination; it is a reminder that any such constraints must be evaluated in part by their behavior under the extreme cases that history periodically generates. Sargent and Wallace [1981] and Sargent [2011] focus on steady-state comparisons; the historical record suggests that the transitions between regimes matter at least as much.

The reflexivity of experience. Each of the reversals was accompanied by an appeal to experience that the original position could not have anticipated. Clay and Madison could not have known in 1791 or 1811 what the War of 1812 would reveal about the fiscal consequences of the absence of a national monetary institution. Friedman could not have known in 1948 what his and Schwartz's subsequent historical research would reveal about the monetary origins of the Great Depression, or what the postwar Keynesian political economy would reveal about the systematic bias toward fiscal expansion. Experience updated the effective parameters of the trade-off identified in Sargent [2011]: the relative magnitude of the fiscal and monetary risks, and the capacity of political institutions to manage them.

The constitutional versus the pragmatic argument. Chase's career and Madison's liquidation doctrine both reveal a tension between constitutional arguments about where institutional lines may be drawn and pragmatic arguments about where they should be drawn. The constitutional argument tries to fix the line against the pressures of the moment; the pragmatic argument allows the line to respond to changing

circumstances. Madison resolved the tension by treating constitutional meaning as itself partly determined by experience (the liquidation doctrine). Chase resolved it by insisting that the Constitution fixed the line absolutely—and discovered that the political economy would not sustain that absolutism.

The ratchet problem. Friedman's final position—the fixed money-growth rule—was motivated in part by his assessment that the coupling of monetary and fiscal policy in a democracy reliably moves in one direction: toward more money creation, more inflation, more fiscal expansion. The historical record broadly supports this concern. Neither Clay nor Madison, having supported the Second Bank, sought to use it as an instrument of fiscal dominance; but Jackson's destruction of the Bank was followed by the pet-bank era, the suspension of 1837, and eventually the Legal Tender Acts. The absence of a durable national monetary institution contributed to a sequence of fiscal-monetary crises that culminated in the most extreme form of fiscal dominance the republic had seen. The ratchet runs toward fiscal dominance when monetary institutions are weak; only strong, credible monetary commitments can reverse it.

The impossibility of personal commitment. Perhaps the deepest lesson of Chase's career is that individuals cannot, through the force of their own convictions, maintain institutional lines against political and fiscal pressures. Chase's personal monetary principles were genuine and deeply held; he had stated them publicly for decades. Under sufficient pressure—the fiscal demands of a war for national survival—he abandoned them. And having abandoned them, he could not restore them from his judicial position, because the political forces that had driven the abandonment had not themselves changed. Institutional lines must be maintained by institutions, not by individuals; and as Madison's liquidation doctrine suggests, institutions themselves are mutable.

9 Conclusion

The trade-off between price-level stability and Ramsey-optimal fiscal efficiency identified in Sargent [2011] is not merely a property of theoretical models. It is a live institutional choice that has generated passionate disagreement, principled reversal, and constitutional crisis throughout American economic history. The four individual cases examined here—Friedman's shift from fiscal coordination to monetary rules;

Clay's conversion from Republican opponent to Whig champion of the national bank; Madison's move from strict-constructionist critic to pragmatic signer of the Second Bank bill; and Chase's extraordinary journey from hard-money Democrat to greenback architect to constitutional counter-revolutionist—all illustrate the same underlying truth: the optimal institutional line responds to circumstances, and circumstances change.

That truth is confirmed, at the legislative rather than the individual level, by the half-century redrawing of the money–credit line examined in the preceding section. The bond-backed currency of the National Banking System was a stable but inflexible answer to the question "what should money be backed by?" The real bills doctrine of the Federal Reserve Act was a more elastic answer that solved the inelasticity problem while importing the procyclical instability that Thornton had identified in 1802 and that Friedman and Schwartz held responsible for the catastrophe of 1929–33. The theoretical reconsideration by Sargent and Wallace [1982], showing that such a regime can be Pareto superior to a quantity-theory alternative, demonstrates that this congressional reversal, like the individual reversals, reflects a genuine analytical trade-off rather than mere confusion. Neither side of the line dominates the other; the right answer depends on the fiscal and regulatory environment.

This is not an argument for institutional nihilism. The historical record also supports Friedman's final conviction that price-level stability requires institutional commitments that are robust to fiscal pressure, and that such robustness is hard to achieve and easy to destroy. The Second Bank was chartered in 1816 in response to the fiscal lessons of 1812 and destroyed by 1836 in response to political pressures that the Bank's institutional designers had not foreseen. The Legal Tender Acts were upheld in *Knox v. Lee* by a Court whose composition had been manipulated for the purpose and confirmed in *Juilliard v. Greenman* by an 8–1 majority that included no defender of Chase's original position. The trajectory is one of progressive erosion of monetary constraints under fiscal pressure.

The cases do counsel, however, for humility about institutional design. Principled people who have thought carefully about where to draw the lines have repeatedly found themselves unable to maintain those lines in the face of changed circumstances, new evidence, and the coercive logic of fiscal emergency. Designing institutions that are both responsive enough to survive such pressures and durable enough to provide

the stability that is their raison d'être is the enduring challenge. The historical reversals documented here are a measure of how difficult that challenge is.

References

George Bancroft. *A Plea for the Constitution of the United States of America Wounded in the House of Its Guardians*. Harper & Brothers, New York, 1886.

Milton Friedman. A monetary and fiscal framework for economic stability. *American Economic Review*, 38(3):245–264, 1948.

Milton Friedman. *A Program for Monetary Stability*. Fordham University Press, New York, 1960.

Milton Friedman and Anna J. Schwartz. *A Monetary History of the United States, 1867–1960*. Princeton University Press, Princeton, 1963.

David Laidler. Misconceptions about the real-bills doctrine: A comment on Sargent and Wallace. *Journal of Political Economy*, 92(1):149–155, 1984.

Jr. Robert E. Lucas and Nancy L. Stokey. Optimal fiscal and monetary policy in an economy without capital. *Journal of Monetary Economics*, 12(1):55–93, 1983.

Thomas J. Sargent. Where to draw lines: Stability versus efficiency. *Economica*, 78 (310):197–214, 2011.

Thomas J. Sargent and Neil Wallace. Some unpleasant monetarist arithmetic. *Federal Reserve Bank of Minneapolis Quarterly Review*, 5(3):1–17, 1981.

Thomas J. Sargent and Neil Wallace. The real-bills doctrine versus the quantity theory: A reconsideration. *Journal of Political Economy*, 90(6):1212–1236, 1982.

Henry Thornton. *An Enquiry into the Nature and Effects of the Paper Credit of Great Britain*. Hatchard, London, 1802.